Jahangir Mammadli

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Education

University of Wisconsin–Madison, Madison, WI — Expected May 2028
Bachelor of Science in Computer Sciences, Data Science Certificate
Transferred from Arizona State University — GPA: 3.79 | Dean's List
Relevant Coursework: Data Structures & Algorithms, Probability & Statistics, Discrete Mathematics, Exploring Data in R & Python, Computer Organization, Programming III (in progress)

Projects

Edwin – Adaptive Learning System (Bayesian Learner Model + LLM Tutor) — edwinlearning.com
Python, Flask, PostgreSQL, Redis, Docker, Gemini / Groq, RAG

- Built a Bayesian learner model tracking per-topic mastery as Beta posteriors over an append-only evidence ledger, with source-weighted observations, exponential recency decay, and information-based confidence replacing a naive variance heuristic; recompute-from-ledger plus DB-level uniqueness makes updates idempotent under retries.
- Implemented adaptive assessment as computerized adaptive testing: selects the next topic by expected variance reduction (Bayesian experimental design), combined with a 10-day-half-life forgetting curve that resurfaces confident-but-stale topics.
- Designed a hierarchical Thompson sampling policy over 7 teaching strategies, shrinking per-student posteriors toward a population posterior and a pedagogical cold-start prior to stay usable at low data; convergence validated on synthetic outcome data.
- Built the retrieval layer over a managed vector store: canonical concept resolver using margin-gated semantic matching, neighbor-chunk expansion, lexical reranking, an abstention gate that labels answers not supported by course materials as general knowledge, and citation remapping validated against the retrieved set.
- Wrote an eval harness reporting coverage, calibration, and per-strategy outcomes across 678 evidence observations (in-sample Brier 0.085, 88% direction accuracy); designed a 30-item human-scored rubric covering retrieval hit, groundedness, and answer quality as the held-out successor.
- Migrated the backend from Snowflake to Postgres (23 tables) and inference to hosted LLM APIs (Gemini, Groq) to cut multi-second latency; Redis sessions with versioned ETag caching. Piloted at ASU in CSE 310 with 42 enrolled students; in discussions for UW–Madison pilots.
- Prepared the system for UW–Madison's cybersecurity risk review: locked down database access, built end-to-end data deletion and erasure tools, retention limits, and privacy controls for FERPA-protected student data.
- **Placed 2nd out of 130+ teams (1,100+ participants) at SunHacks 2025.**

SAFE-POS – Anomaly Detection for Point-of-Sale Fraud — *FastAPI, PostgreSQL, React*

- Engineered a POS system detecting cashier fraud via transaction and shift-log pattern analysis; designed backend services and schemas for anomaly detection, with a restaurant pilot in preparation.

Nebulo – Student Tech Collaboration Platform — *React, Node.js, Express.js, MySQL, AWS*

- Built a full-stack platform for student team formation and project launch; designed RESTful APIs deployed on AWS RDS and S3.

Experience

Backend Developer Intern – Yolgen (AI-Driven Professional Insights Platform) — Remote — June 2025 – August 2025

- Developed backend APIs and database models powering AI-driven professional simulation features.
- Integrated OpenAI GPT APIs to generate real-time career feedback and adaptive recommendations.
- Implemented authentication, session management, and data pipelines for personalized insights.
- Reduced median API response time ˜30% through query optimization and caching, working in a 5-person agile team.

Leadership

Founder & Technical Lead – Edwin

- Own the full ML and systems architecture: learner modeling, policy layer, retrieval, evaluation, and deployment.
- Presented to ASU's AI team and department head; advised by UW–Madison's Associate Vice Chancellor for Entrepreneurship on institutional pilots.

Technical Skills

Machine Learning: Bayesian inference, Thompson sampling / multi-armed bandits, adaptive experimental design, model calibration and evaluation, RAG, embeddings, PyTorch, HuggingFace
Languages: Python, SQL, TypeScript, JavaScript (ES6+), C, C++, Java
Backend & Data: Flask, FastAPI, Node.js, PostgreSQL, Redis, Snowflake, MySQL, Pandas, NumPy
Frontend & Infrastructure: React, Next.js, Docker, AWS (RDS, S3), Git/GitHub, Linux, gunicorn, REST APIs